

Institut universitaire des sciences

Faculte des sciences et des technologies

TD N°8 – Réseaux I

Nom & Prénom : Leamande Evena

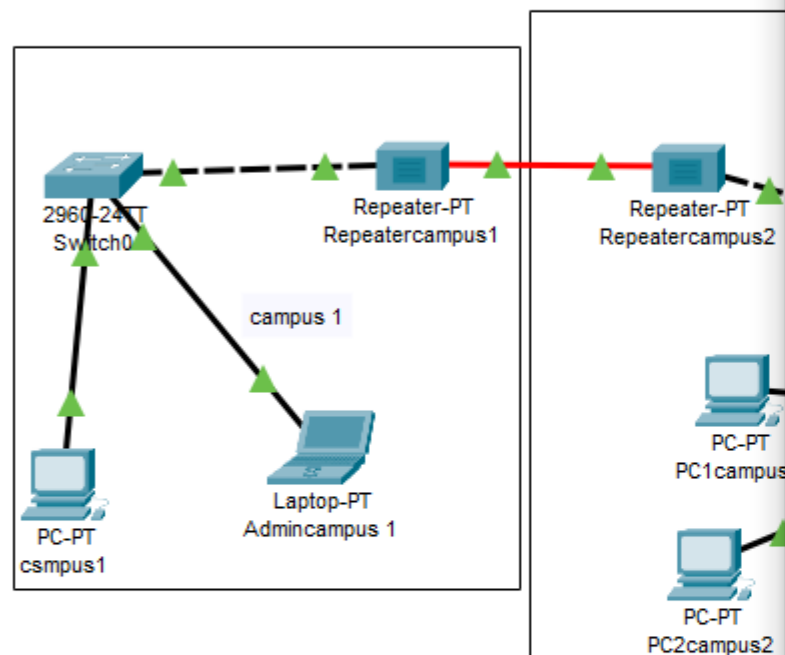
Niveau : L3/FST

Date : le/03/01/2

Objectif de ce TD : était de mettre en place et de sécuriser le réseau, Segmenter le réseau en créant des VLANs pour les services Ventes, Marketing et IT, Automatiser la gestion des VLANs sur tous les commutateurs grâce au protocole **VTP**.



Logical Physical x: 538, y: 143



Time: 02:43:37



Multilayer Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/4 VL

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/4 on VLAN0001. Inconsisten
type.

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state

SWP-campus2>en
SWP-campus2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SWP-campus2(config)#vtp domain ius.com
Changing VTP domain name from NULL to ius.com
SWP-campus2(config)#vtp password 0000
Setting device VLAN database password to 0000
SWP-campus2(config)#vtp mode server
Device mode already VTP SERVER.
SWP-campus2(config)#vtp version 2
SWP-campus2(config)#spanning-tree mode rapid-pvst
SWP-campus2(config)#spanning-tree vlan 10 root primary
SWP-campus2(config)#spanning-tree vlan 20 root secondary
SWP-campus2(config)#int f0/2
SWP-campus2(config-if)#switchport port-security
Command rejected: FastEthernet0/2 is a dynamic port.
SWP-campus2(config-if)#switchport mode access
SWP-campus2(config-if)#switchport port-security
SWP-campus2(config-if)#switchport port-security maximum 3
SWP-campus2(config-if)#switchport port-security violation restrict
SWP-campus2(config-if)#exit
SWP-campus2(config)#
```

Copy

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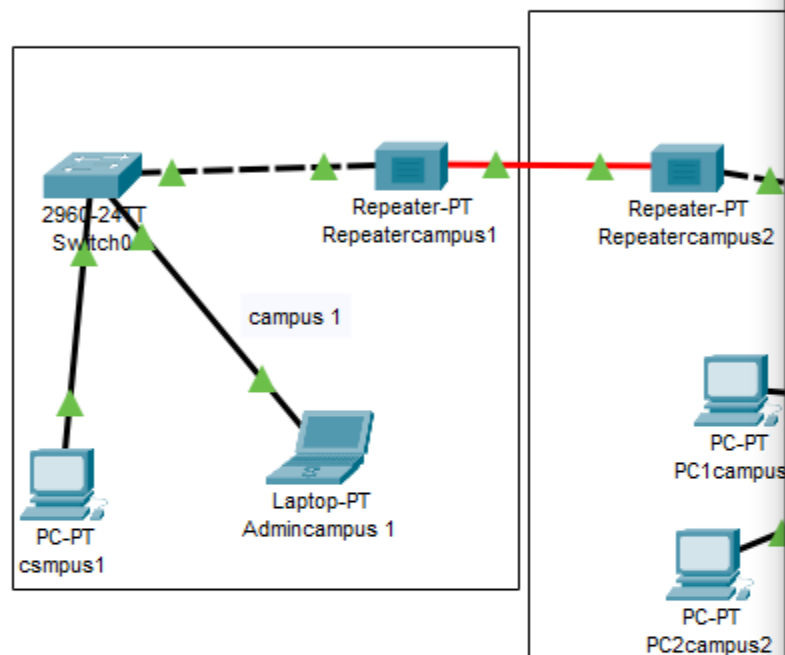


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Logical Physical x: 889, y: 90



Time: 02:18:50



Physical Config CLI Attributes

IOS Command Line Interface

```
SW-campus2>en
SW-campus2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW-campus2(config)#int f0/1
SW-campus2(config-if)#switchport mode trunk

SW-campus2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed sta

SW-campus2(config-if)#exit
SW-campus2(config)#int range f0/4-9
SW-campus2(config-if-range)#switchport mode access
SW-campus2(config-if-range)#switchport access vlan 10
SW-campus2(config-if-range)#exit
SW-campus2(config)#int range f0/10-17
SW-campus2(config-if-range)#switchport mode access
SW-campus2(config-if-range)#switchport access vlan 20
SW-campus2(config-if-range)#exit
SW-campus2(config)#int range f0/18-24
SW-campus2(config-if-range)#switchport mode access
SW-campus2(config-if-range)#switchport access vlan 30
SW-campus2(config-if-range)#exit
SW-campus2(config)#
```

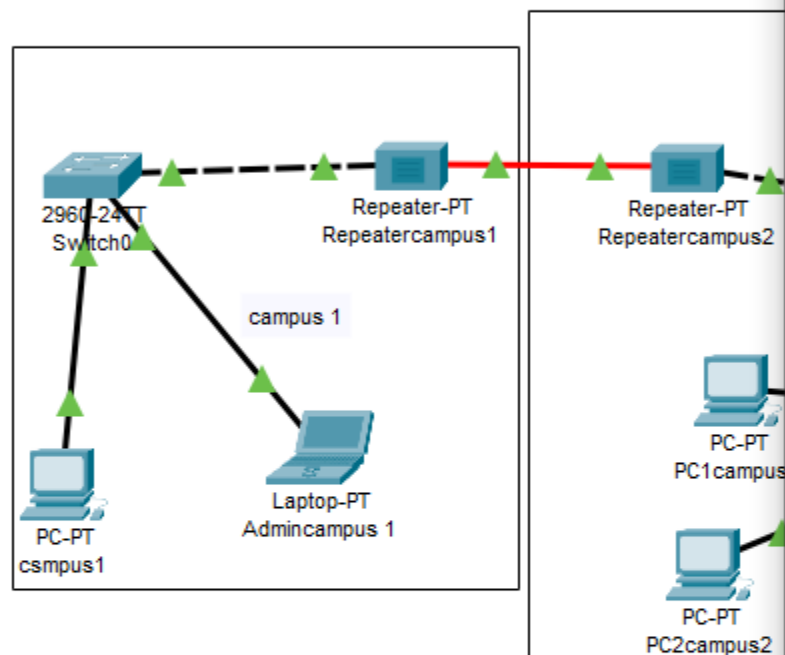
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Logical Physical x: 542, y: 125



Time: 02:18:42



Multilayer Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SWP-campus2
SWP-campus2(config)#vlan 10
SWP-campus2(config-vlan)#name ventes
SWP-campus2(config-vlan)#vlan 20
SWP-campus2(config-vlan)#marketing
^
% Invalid input detected at '^' marker.

SWP-campus2(config-vlan)#name marketing
SWP-campus2(config-vlan)#vlan 30
SWP-campus2(config-vlan)#name IT
SWP-campus2(config-vlan)#exit
SWP-campus2(config)#int g0/1
SWP-campus2(config-if)#switchport mode access
SWP-campus2(config-if)#switchport mode trunk
SWP-campus2(config-if)#int g0/2
SWP-campus2(config-if)#switchport mode access
SWP-campus2(config-if)#switchport mode trunk
SWP-campus2(config-if)#exit
SWP-campus2(config)#int f0/24
SWP-campus2(config-if)#switchport mode access
SWP-campus2(config-if)#switchport mode trunk
SWP-campus2(config-if)#exit
SWP-campus2(config)%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non t
FastEthernet0/1 VLAN1.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/1 on VLAN0001. Inconsis
type.
```

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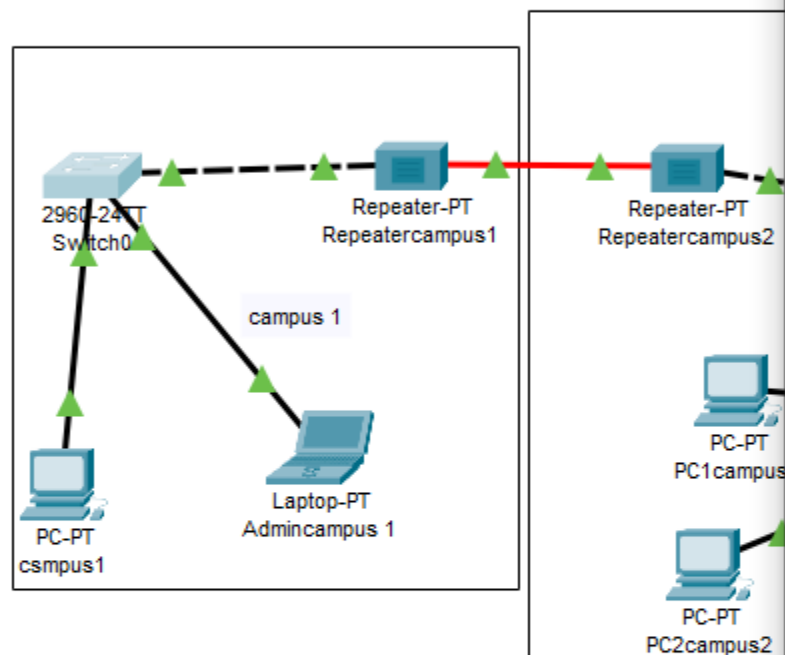


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Logical Physical x: 1105, y: 5



Time: 02:18:26



Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed sta
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed sta
```

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname campus1
campus1(config)#vlan 10
campus1(config-vlan)#name ventes
campus1(config-vlan)#vlan 20
campus1(config-vlan)#name marketing
campus1(config-vlan)#vlan 30
campus1(config-vlan)#name IT
campus1(config-vlan)#exit
campus1(config)#
```

campus1 con0 is now available

Press RETURN to get started.

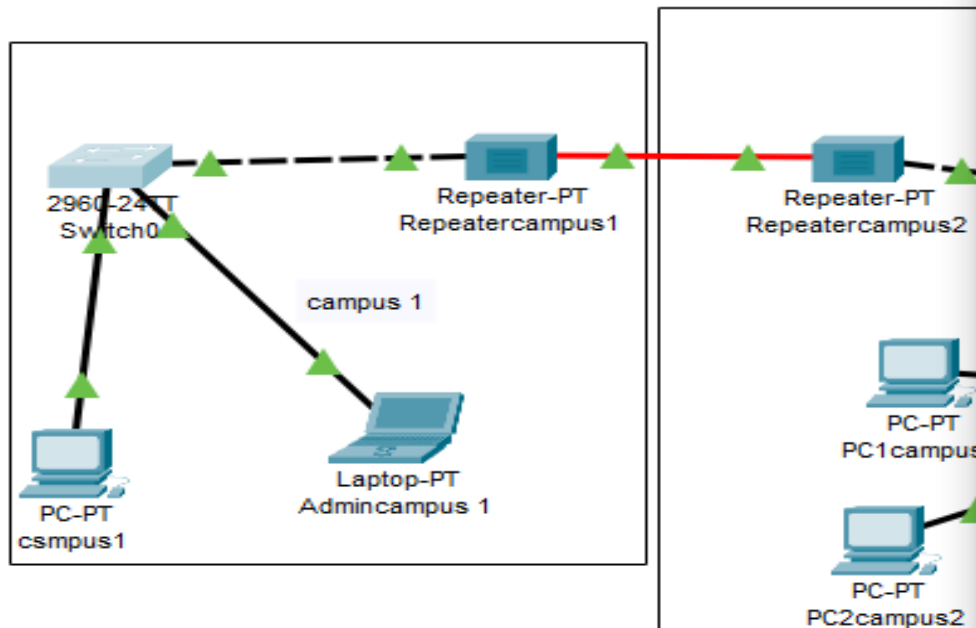
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Logical Physical x: 151, y: 205



Time: 02:18:12



```
campus1>en
campus1#conf t
Enter configuration commands, one per line. End with Ctrl-Z to exit.
campus1(config)#int f0/3
campus1(config-if)#switchport mode trunk

campus1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface

campus1(config-if)#exit
campus1(config)#int range f0/4-9
campus1(config-if-range)#switchport mode access
campus1(config-if-range)#switchport access vlan
campus1(config-if-range)#exit
campus1(config)#int range f0/10-17
campus1(config-if-range)#switchport mode access
campus1(config-if-range)#switchport access vlan
campus1(config-if-range)#exit
campus1(config)#int range f0/18-24
campus1(config-if-range)#switchport mode access
campus1(config-if-range)#switchport access vlan
campus1(config-if-range)#exit
campus1(config)#
```

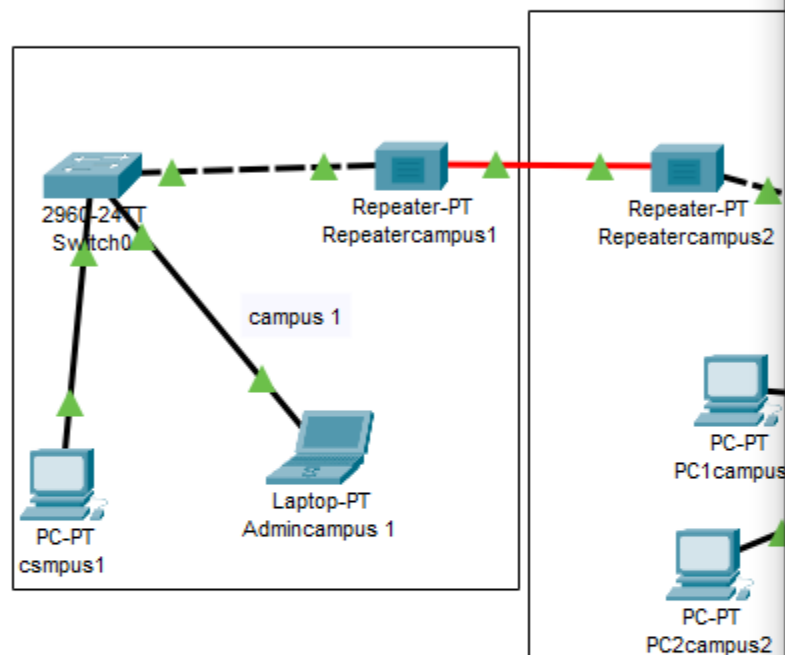
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Logical Physical x: 538, y: 143



Time: 02:44:46



Physical Config CLI Attributes

IOS Command Line Interface

```
Setting device VLAN database password to 0000
SWP-campus2(config)#vtp mode server
Device mode already VTP SERVER.
SWP-campus2(config)#vtp version 2
SWP-campus2(config)#spanning-tree mode rapid-pvst
SWP-campus2(config)#spanning-tree vlan 10 root primary
SWP-campus2(config)#spanning-tree vlan 20 root secondary
SWP-campus2(config)#int f0/2
SWP-campus2(config-if)#switchport port-security
Command rejected: FastEthernet0/2 is a dynamic port.
SWP-campus2(config-if)#switchport mode access
SWP-campus2(config-if)#switchport port-security
SWP-campus2(config-if)#switchport port-security maximum 3
SWP-campus2(config-if)#switchport port-security violation restrict
SWP-campus2(config-if)#exit
SWP-campus2(config)#exit
SWP-campus2#
%SYS-5-CONFIG_I: Configured from console by console
```

SWP-campus2#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/3, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	ventes	active	
20	marketing	active	
30	IT	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

SWP-campus2#

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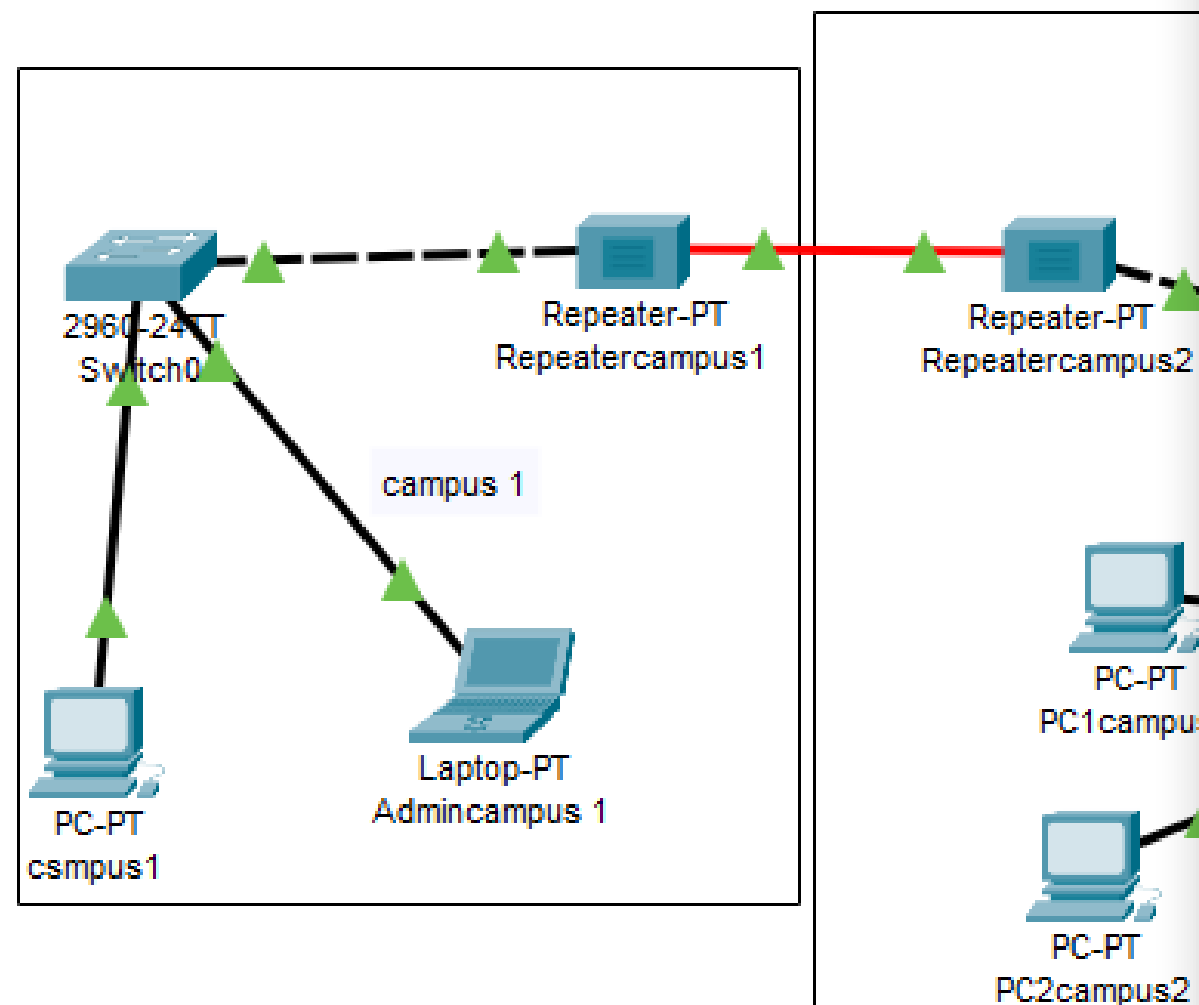


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Logical Physical x: 538, y: 143



```
1 default

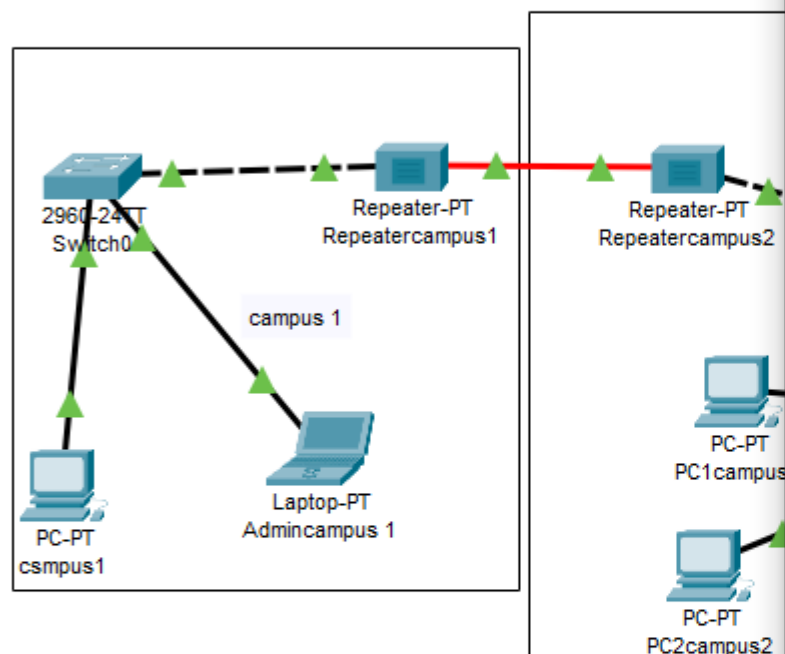
10 ventes
20 marketing
30 IT
1002 fddi-default
1003 token-ring-default
1004 fddinet-default
1005 trnet-default
SWP-campus2#show vlan
```

VLAN	Name
1	default

```
10 ventes
20 marketing
30 IT
1002 fddi-default
1003 token-ring-default
1004 fddinet-default
1005 trnet-default
```



Logical Physical x: 538, y: 143



Time: 02:46:43



Physical Config CLI Attributes

IOS Command Line Interface

```

Fa0/7, Fa0/8, Fa0/9, Fa0/10
Fa0/11, Fa0/12, Fa0/13, Fa0/14
Fa0/15, Fa0/16, Fa0/17, Fa0/18
Fa0/19, Fa0/20, Fa0/21, Fa0/22
Fa0/23, Fa0/24, Gig0/1, Gig0/2

10  ventes                active
20  marketing            active
30  IT                   active
1002 fddi-default        active
1003 token-ring-default  active
1004 fddinet-default     active
1005 trnet-default       active

VLAN Type  SAID      MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
-----
1    enet   100001    1500  -      -      -      -   -        0      0
10   enet   100010    1500  -      -      -      -   -        0      0
20   enet   100020    1500  -      -      -      -   -        0      0

SWP-campus2#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     auto      n-802.1q       trunking    1
Fa0/4     auto      n-802.1q       trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005
Fa0/4     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30
Fa0/4     1,10,20,30

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30
Fa0/4     1,10,20,30

SWP-campus2#

```

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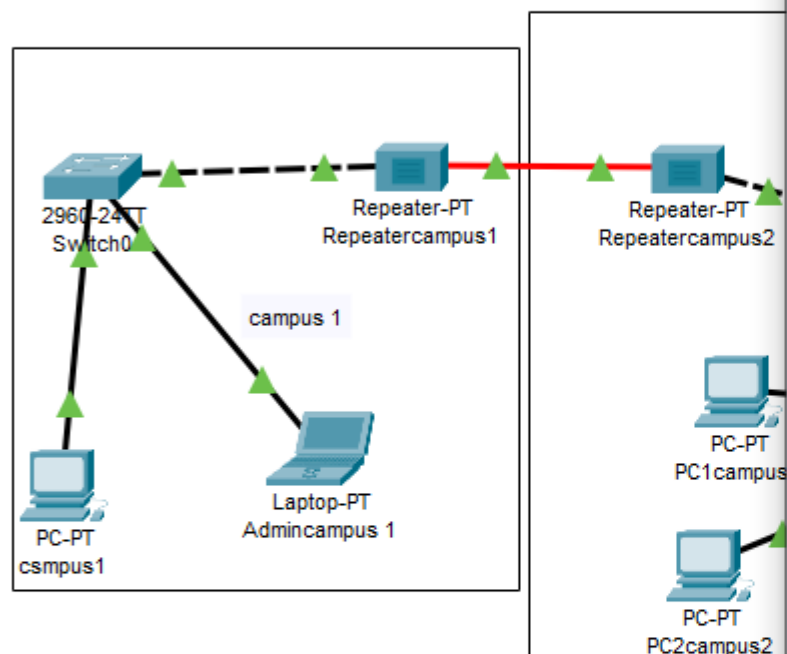
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Logical Physical x: 538, y: 143



Time: 02:47:21



Multilayer Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
SWP-campus2#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     auto      n-802.1q       trunking    1
Fa0/4     auto      n-802.1q       trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005
Fa0/4     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30
Fa0/4     1,10,20,30

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30
Fa0/4     1,10,20,30

SWP-campus2#show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          : ius.com
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 000D.BDE0.3200
Configuration last modified by 0.0.0.0 at 3-1-93 02:27:45
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN :
-----
VTP Operating Mode       : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 8
Configuration Revision   : 7
MD5 digest               : 0x68 0x0A 0xA6 0x01 0xEE 0x26 0x16 0x9D
                          : 0x14 0x33 0xD1 0xDC 0x5C 0x6C 0xB0 0x76

SWP-campus2#
```

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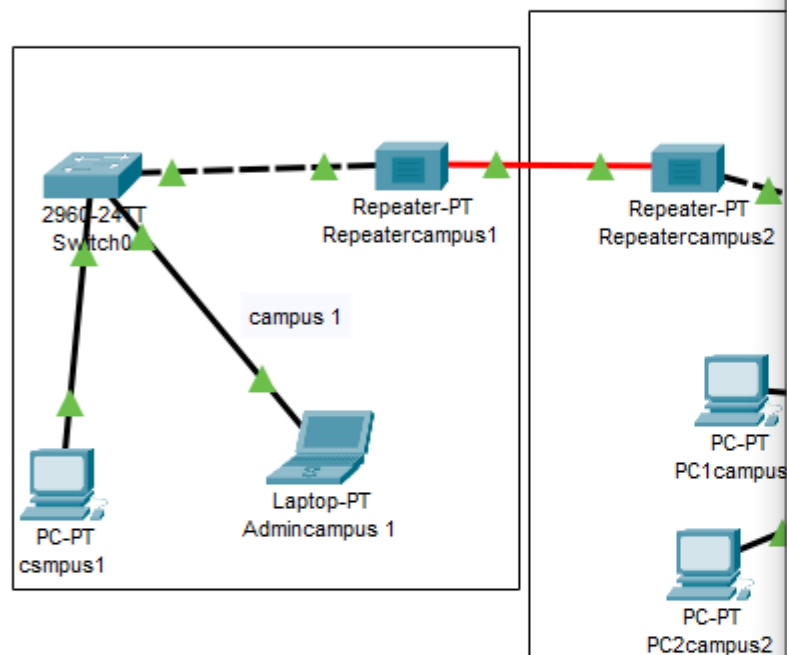


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Logical Physical x: 538, y: 143



Time: 02:48:55



Multilayer Switch0

Physical Config CLI Attributes

IOS Command Line Interface

Feature VLAN :

```

-----
VTP Operating Mode           : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs      : 8
Configuration Revision        : 7
MDS digest                   : 0x68 0x0A 0xA6 0x01 0xEE 0x26 0x16 0x9D
                               0x14 0x33 0xD1 0xDC 0x5C 0x6C 0xB0 0x76
  
```

SWP-campus2#show spanning-tree security

```

^
% Invalid input detected at '^' marker.
  
```

SWP-campus2#show spanning-tree summary

```

Switch is in rapid-pvst mode
Root bridge for: ventes marketing
Extended system ID           is enabled
Portfast Default              is disabled
PortFast BPDU Guard Default  is disabled
PortFast BPDU Filter Default is disabled
Loopguard Default             is disabled
EtherChannel misconfig guard is disabled
UplinkFast                    is disabled
BackboneFast                  is disabled
Configured Pathcost method used is short
  
```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	1	0	0	3	4
VLAN0010	3	0	0	1	4
VLAN0020	3	0	0	1	4
VLAN0030	3	0	0	1	4
4 vlans	10	0	0	6	16

SWP-campus2#

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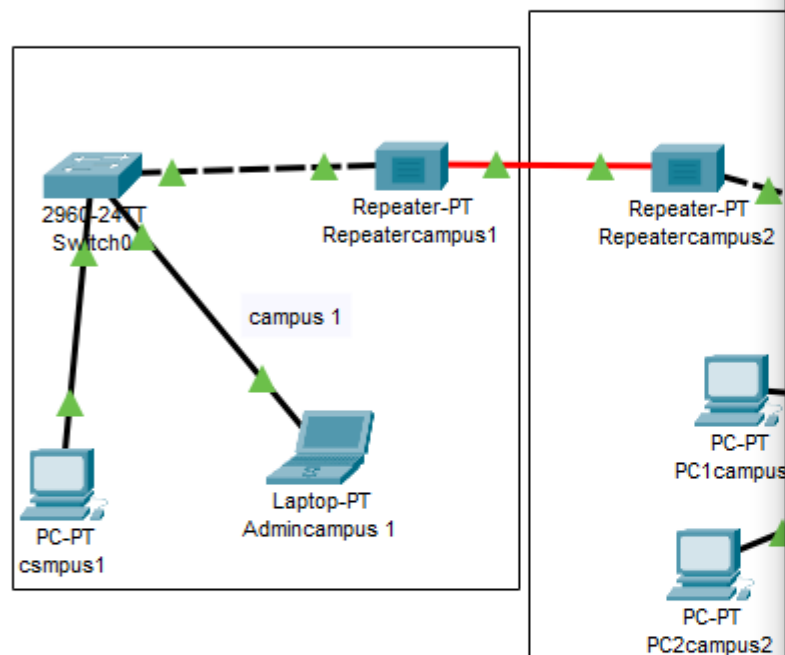
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Logical Physical x: 619, y: 56



Time: 02:50:29



Router1

Physical Config CLI Attributes

IOS Command Line Interface

campus2 con0 is now available

Press RETURN to get started.

```
campus2>en
campus2#ping 192.168.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

campus2#
```

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Pour conclure, j'ai réussi à configurer une infrastructure réseau complète et sécurisée pour les deux campus.les informations de VLAN sont synchronisées partout. Le réseau est désormais protégé contre les erreurs de boucle et les accès non autorisés. Les tests effectués montrent que les liaisons **Trunk** fonctionnent parfaitement, ce qui rend le réseau prêt pour une utilisation réelle.